

SUNSPOT NUMBER EFFECT ON EARTH'S CLIMATE CHANGE AND WATER SUSTAINABILITY FOR LONG ISLAND, NY

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Outline

Introduction

Data

- We use daily sunspot number data measured from 1700 to 2021 processed in R programming language.
- Temperature data derived from:
- -NASA

Methodology

- Application of the moving average filter on sunspot numbers.
- Application of the KZ filter to remove the short-term variations from the time series and reveal the long-term trend
- Trend pattern observation between the long term of the daily change of sunspot number and temperature.

Kolmogorov-Zurbenko Filter ($KZ_{m,p}$)

- For the decomposition of the time series we use the Kolmogorov-Zurbenko filter. The filter is defined by:

$$Y_t = \frac{1}{m} \sum_{j=-k}^k X_{t+j}$$

$$m=2k+1$$

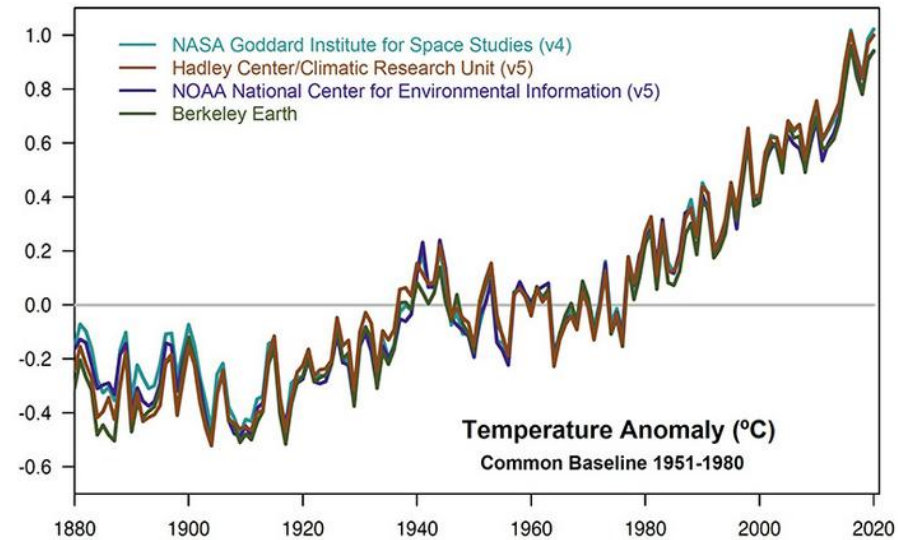
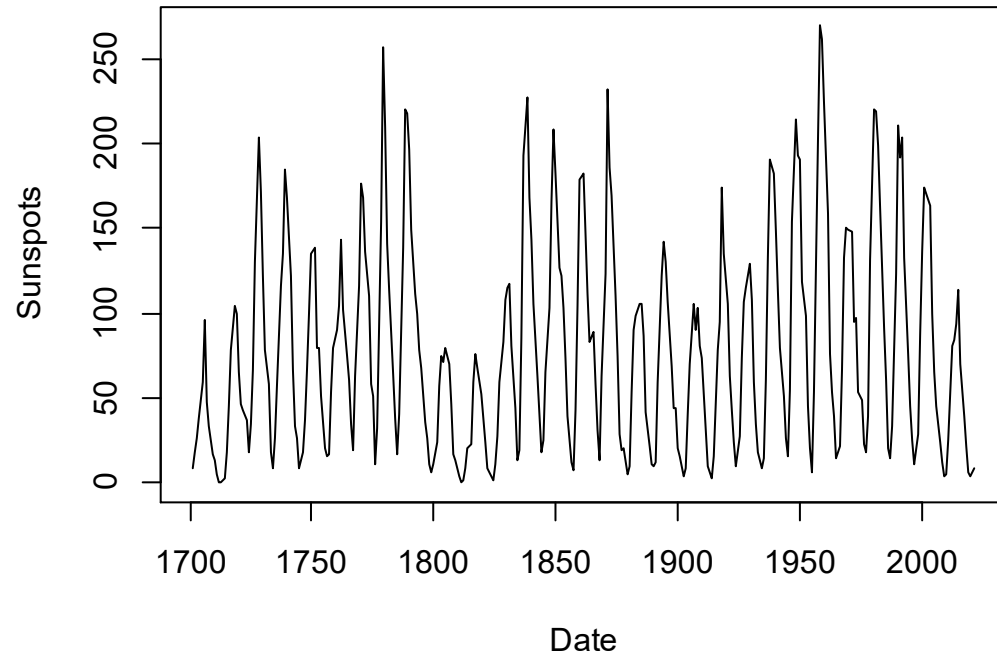
X_t : original series

Y_t : filtered series

p : number of iterations

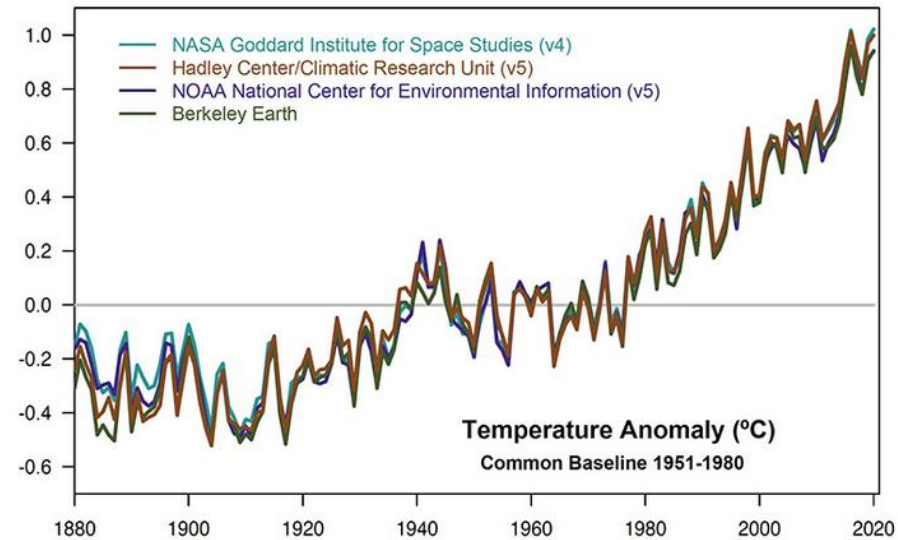
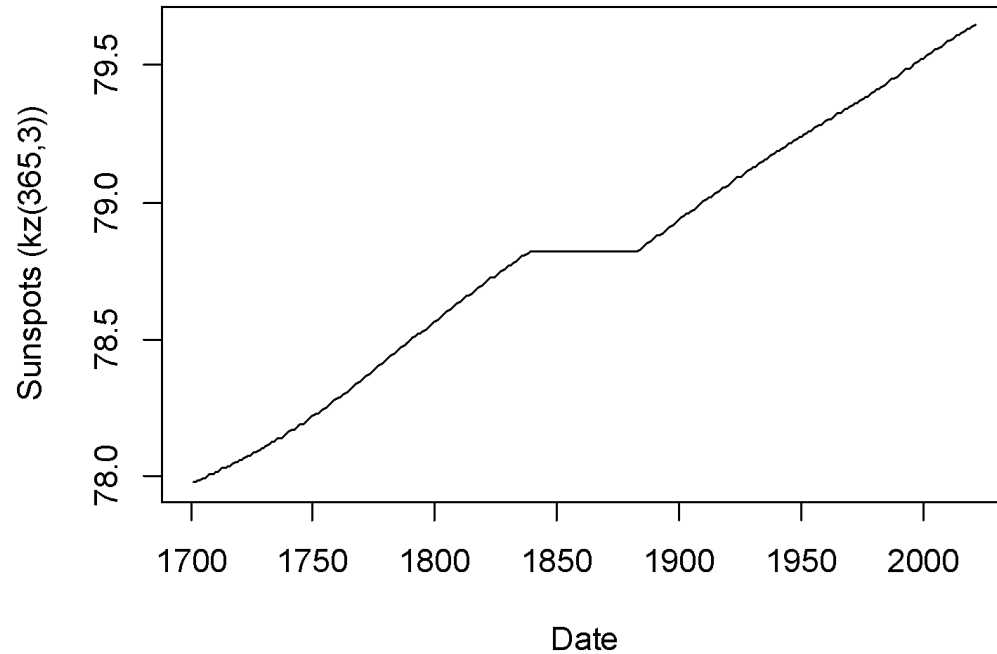
Annual data of Sunspots compared with climate change from 1700 to 2021

ANNUAL DATA

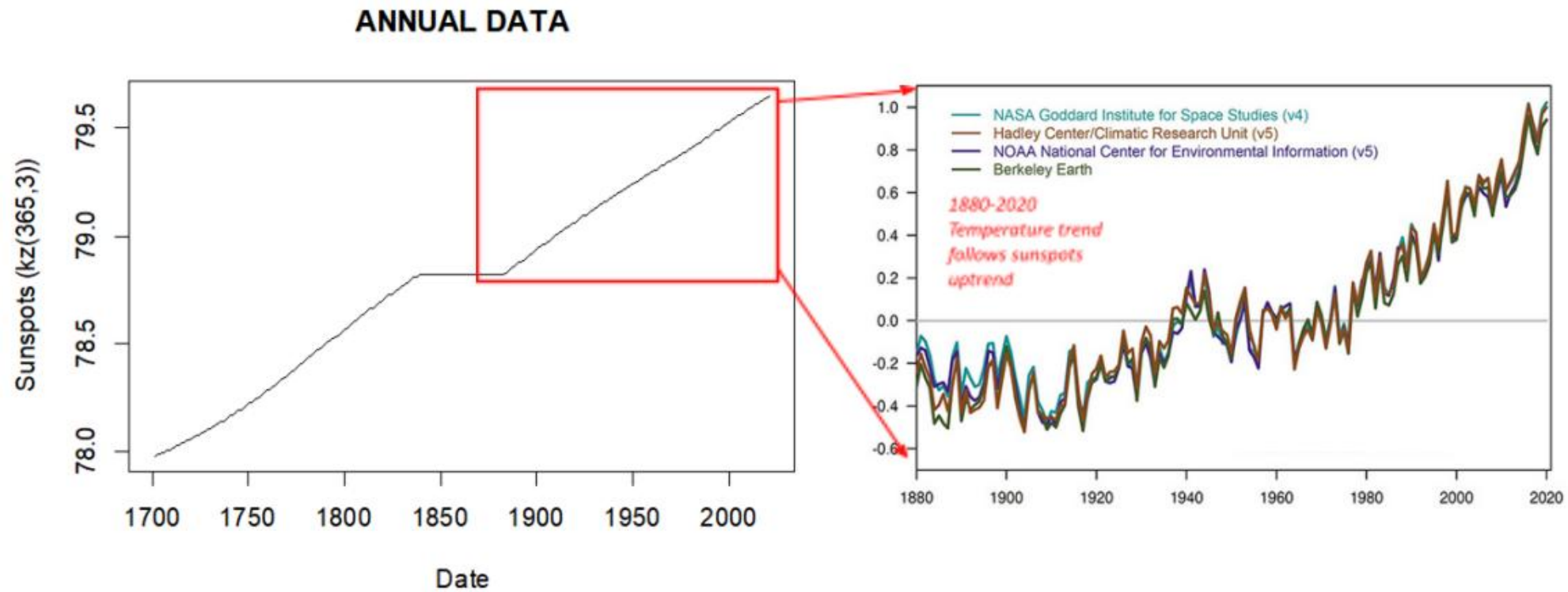


Long term component of Sunspots compared with climate change from 1700 to 2021

ANNUAL DATA



Long term component of Sunspots compared with climate change from 1700 to 2021



Conclusions

- Select key sentences or words from your conclusion section of your paper

References

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Acknowledgement

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